

SleepFM-Unify Paper Draft

SleepFM-Unify: Robust Multimodal Pretraining under Heterogeneous and Missing PSG

****Draft manuscript (methods paper)****

****Status:**** engineering-complete codebase + synthetic demos; ****CinC / SHHS / MESA quantitative claims 待补充****

****Target framing:**** Nature Machine Intelligence-style methods article / ICML-compatible experimental spine

****nature-skills:**** `nature-writing` (methods + nat-mach-intell), `nature-figure` (Python / SciencePlots)

****Code:**** <https://github.com/Coucou2016/SleepFM-Unify> (`main`)

Abstract

Polysomnography (PSG) is routinely incomplete: montages differ across clinics, leads drop out overnight, and public corpora expose mismatched channel sets. Sleep foundation models such as SleepFM align brain, cardiac and respiratory streams with leave-one-out (LOO) contrastive learning, but pure cross-modal alignment under-specifies behaviour when modalities or channels are missing. We introduce ****SleepFM-Unify****, a robustness layer on SleepFM 1D EfficientNet encoders for ****heterogeneous and arbitrarily missing PSG****: sample-wise modality dropout, `present\_mask`-aware encoding (no BatchNorm pollution), a missingness InfoNCE term gated by original presence, optional per-lead `channel\_mask` zeroing, and mixed LOO/pairwise objectives on a shared subspace with private capacity retained for downstream probes. Shared-private heads are an implementation tool (FOCAL already factorizes shared/private multimodal time series); ****they are not our novelty claim****. We release configs, export/validation gates and a dual LOO-Unify paper suite. ****Quantitative CinC/SHHS/MESA results remain 待补充**** pending user DUA downloads; synthetic demos are engineering smoke tests (AUROC near chance) and are not scientific claims.

Introduction

Sleep staging and sleep-disordered breathing (SDB) assessment rely on multimodal PSG. SleepFM showed that large-scale LOO contrastive pretraining across brain-activity (BAS), ECG and respiratory streams yields transferable embeddings for staging, SDB detection and cross-modal retrieval. Follow-on models emphasize scale, structure and incomplete montages: a Nature Medicine SleepFM line scales LOO-CL to disease-risk prediction with SHHS transfer; PhysioOmni-style and Omni-Sleep lines impose physiological hierarchies; CIMSleeNet targets incomplete multimodal staging via modal imagination; OSF/SleepBench-style benchmarks stress standardized evaluation under montage heterogeneity.

A remaining systems gap is ****SleepFM-compatible training and evaluation that remains well-defined when entire modalities or individual leads are absent****, without replacing the paper LOO baseline (`unify.enabled: false`). SleepFM-Unify addresses this with mask-aware encoding, missingness objectives and evaluation honesty gates—not by claiming a new shared-private factorization theory.

****Contributions (evidence-gated):****

1. Heterogeneous / missing-PSG training stack on SleepFM encoders: sample-wise modality dropout, mask-aware BN skip, $\mathcal{L}_{\text{miss}}$ gated by original presence, optional lead masks.
2. Shared-private projection heads as a **tool** (mixed L00 + pairwise on shared; private for concat / space probes), with orthogonality + VICReg-style private hinge—explicitly positioned as FOCAL-adjacent, not novel factorization.
3. Evaluation honesty gates: channel meta fail-fast, CinC label coverage, apnea-epoch-rate vs clinical AHI wording, seeded RNG gallery caps, strict split isolation (`RuntimeError` on leak), few-shot mean $\pm$ 95% CI (≥ 10 participant-level repeats in paper mode).
4. Reproducible CLI paper suite for L00 vs Unify comparisons once real exports exist (shared/private/concat probes included).

Non-claims. Synthetic AUROC $\approx$ 0.5 is demo-only. Night `ahi_bin` uses apnea-epoch-rate cut-points, not clinical AASM AHI. Nature Medicine disease C-Index values are not our local results. We do **not** claim novelty for shared-private factorization per se.

Related work

Sleep foundation models. SleepFM (Thapa et al., ICML 2024; PMLR 235:48019–48037; arXiv:2405.17766) introduced multimodal leave-one-out contrastive learning (L00-CL) on BAS/ECG/respiratory PSG clips and showed L00 outperforming pairwise alignment on staging, SDB detection and cross-modal retrieval (e.g. reported macro AUROC 0.88 vs CNN 0.72 for staging in that paper — **prior art only**). A later SleepFM line (Nature Medicine, doi:10.1038/s41591-025-04133-4; $\sim$ 585k hours / $\sim$ 65k participants; disease C-Index reporting) scales L00-CL to future disease risk and SHHS transfer — **do not conflate** those clinical numbers with the ICML 2024 staging/retrieval setting, and do not paste them into our CinC/SHHS tables.

Incomplete multimodal sleep. CIMSsleepNet (NeurIPS 2024) targets incomplete multimodal staging via modal imagination (MAIM) plus semantic/modal calibration contrastive learning. SleepFM-Unify instead keeps SleepFM’s L00 interface and trains under dropout/masks without an imagination decoder.

Physiological hierarchy / omnibus models. Omni-Sleep (arXiv:2607.07720) uses CNS/ANS partitions with intra-/inter-system contrastive terms. PhysioOmni-style work similarly emphasizes structured physiological priors across biosignals. Unify does not introduce a new ontology; it hardens SleepFM under missingness.

Shared-private multimodal learning. FOCAL and related shared-private factorizations for multimodal time series already separate aligned and modality-specific subspaces (often with orthogonality). Unify **instantiates** shared/private heads inside SleepFM’s encoder interface so contrastive terms can stay on shared embeddings while private capacity remains for downstream concat and missing-modality regimes. **Novelty** is not “we invented shared-private.”

Benchmarks. OSF / SleepBench-style efforts standardize sleep foundation evaluation under heterogeneous setups. Our contribution is a reproducible SleepFM-compatible training/eval stack with honesty gates; we do not replace SleepBench.

Positioning (one sentence). Relative to SleepFM (flat L00), Nature Med SleepFM (scale + disease risk), CIMSsleepNet (imagination under missingness), PhysioOmni/Omni-Sleep (physiological hierarchy), FOCAL (shared/private factorization), and OSF/SleepBench (benchmarking), SleepFM-Unify is a **lightweight** missing/heterogeneous-PSG robustness layer on

SleepFM-compatible encoders**, designed for fair L00-vs-Unify ablations once real PSG exports exist.

Method

Backbone (unchanged SleepFM)

Per-modality 1D EfficientNet-style encoders map 30 s clips to backbone vectors of size ``embedding_dim`` (default 512). Baseline contrastive mode is leave-one-out InfoNCE (``configs/default.yaml`, `unify.enabled: false``).

Missing / heterogeneous PSG stack

- **Modality dropout:** default sample-wise random non-empty subsets (``modality_dropout_mode: sample``); batch drop-one remains available.
- **Mask-aware encode:** rows with ``present_mask=0`` skip the encoder (no BatchNorm pollution); missing-modality embeddings stay exactly zero.
- **$\mathcal{L}_{\mathrm{miss}}$:** InfoNCE(remaining shared mean, dropped shared) only where the dropped modality was **originally present**.
- **Channel mask (partial):** dataset $\rightarrow$ collate $\rightarrow$ encode zeros padded/absent leads before the CNN. **Done:** lead zeroing + fixed montage schema (10/2/7 with documented pads).
- TODO:** true variable-channel encoders / mask-aware channel attention beyond zero-fill.

Shared - private heads (tool, not claim)

When Unify is enabled, each backbone vector is mapped by linear heads:

$$\begin{aligned} & \backslash[\\ z_m &= [z_m^{\mathrm{shared}}; z_m^{\mathrm{private}}], \quad \backslash\quad \\ & \dim(z_m^{\mathrm{shared}}) = \dim(z_m^{\mathrm{private}}) = 256 \\ & \backslash] \end{aligned}$$

by default, preserving a 512-d concat for downstream logistic heads comparable to SleepFM. Contrastive terms act on shared only; private stays for concat / ``shared|private`` space probes. Orthogonality: center columns, column-L2 normalize, then $\mathbf{mean}((S^{\mathrm{top}} P)^2)$. Optional VICReg-style private variance hinge. Optional temporal head (GRU/Transformer) on shared epoch sequences.

Mixed objective

$$\begin{aligned} & \backslash[\\ \mathcal{L} &= \lambda_{\mathrm{L00}} \mathcal{L}_{\mathrm{L00}} \\ & + \lambda_{\mathrm{pair}} \mathcal{L}_{\mathrm{pair}} \\ & + \lambda_{\mathrm{priv}} \mathcal{L}_{\mathrm{private}} \\ & + \lambda_{\mathrm{orth}} \mathcal{L}_{\mathrm{orth}} \\ & + \lambda_{\mathrm{miss}} \mathcal{L}_{\mathrm{miss}} \\ & + \lambda_{\mathrm{temp}} \mathcal{L}_{\mathrm{temp}} \\ & \backslash] \end{aligned}$$

Default Unify weights: see ``configs/unify.yaml`` and ``docs/UNIFY.md``.

Downstream and retrieval

Default downstream space concatenates shared $\backslash$ private per modality. CLI / paper suite also probe **shared-only** and **private-only**. Retrieval uses **shared** embeddings from the same checkpoint. Gallery caps use seeded RNG subsample (`limit_gallery(..., mode="rng")``), not a loader-order prefix.

Few-shot protocol

Participant-level k -shot linear probes with **≥ 10 repeats** in paper mode (`scripts/eval_fewshot.py``, `run_paper_suite.py``); report **mean $\pm$ 95% CI** (Student- t on repeat means). Demo/CI may use 2 repeats.

Honesty gates (part of the method interface)

- Channel meta fail-fast when official 5/1/3 montage weights meet schema 10/2/7 without explicit override.
- CinC label-coverage gate blocks degenerate staging/SDB claims when annotations are arousal-only.
- Night probes report **apnea_epoch_rate**, never clinical AHI unless a true AHI column is supplied.
- Split isolation: all pretrain/valid/train/test pairs disjoint on path, participant, and night/recording ids — leaks raise `RuntimeError``.

Experiments

Datasets (access-gated)

Dataset	Role	Status in this draft
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Synthetic demo	CI / smoke	Present (<code>data/synthetic`</code>)
CinC 2018 fixture	Schema export test	Present (<code>data/cinc2018_fixture`</code>)
PhysioNet CinC 2018	Paper pretrain/eval	待补充 (DUA download; see <code>docs/DATA_ACCESS.md`</code>)
NSRR SHHS / MESA	Transfer / night labels	待补充 (DUA download)

Protocol (when data land)

1. `check_data_ready --stage raw``  $\rightarrow$  export  $\rightarrow$  `check_data_ready --stage pretrain`` $\rightarrow$ validate (strict isolation).
2. Pretrain L00 baseline vs Unify (`configs/default.yaml`` vs `configs/unify.yaml``).
3. Downstream staging / apnea probes (concat / shared / private), retrieval Recall@k, modality ablation, few-shot (≥ 10 repeats, mean $\pm$ 95% CI), night-level probes.
4. Optional temporal: `configs/unify_temporal.yaml`` + `evaluate_night.py``.
5. Multi-seed uncertainty on real data (planned; **待补充**).

Experiment matrix (planned)

Experiment	Baseline	Unify	Metric	Status
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Staging (macro AUROC/AUPRC)	L00	Full	Macro AUROC	**待补充**
SDB / apnea probe	L00	Full	AUROC	**待补充**
Retrieval Recall@1/@5	L00	Shared	Recall@k	**待补充**
Modality dropout robustness	L00	Full $\pm$ miss	Δ AUROC	**待补充**
Space probe (concat / shared / private)	—	Unify	Macro AUROC	**待补充**
Ablation orth / miss / +temporal	—	Variants	Same	**待补充**
Few-shot ($k \in \{1, 2, 4\}$, ≥ 10 repeats)	L00	Full	Macro AUROC mean $\pm$ 95% CI	**待补充**
Night κ / apnea_epoch_rate bin	L00 pool	+temporal	κ , AUROC	**待补充**

Results

Table 1. Synthetic demo smoke metrics (NOT paper claims)

Setting	Note	Staging macro AUROC	Retrieval
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Synthetic L00 / Unify	Labels random-ish	≈ 0.5 (chance)	Demo only
CinC 2018	Real PSG	**待补充**	**待补充**
SHHS	Real PSG	**待补充**	**待补充**

Loss curves in Fig. 2 are from a **local few-epoch Unify pretrain on synthetic data** (engineering verification). Orthogonality Gram heatmaps (Fig. 4) are forward-pass diagnostics on the same demo checkpoint. Ablation bars in Fig. 3 and modality-dropout curves in Fig. 5 are **schematic / chance-level** and must not be cited as method superiority.

Ablations (planned; numbers 待补充 on real data)

L00 baseline; Unify full; Unify orth; Unify miss; Unify + temporal; space probes; modality dropout robustness.

Discussion

Unify keeps SleepFM’s L00 retrieval semantics in the shared space while retaining private capacity for downstream concat, and makes missing-modality / heterogeneous-montage training first-class. Honesty constraints (label gates, AHI wording, channel meta, strict isolation) are part of the scientific interface.

Failure modes. (i) Montage mismatch without override $\rightarrow$ load fails by design. (ii) CinC staging claims without coverage $\rightarrow$ gate blocks. (iii) Interpreting synthetic AUROC ≈ 0.5 as a positive result $\rightarrow$ rejected by caption policy. (iv) Calling night apnea-epoch-rate “AHI” $\rightarrow$ wording violation. (v) Claiming novel shared-private factorization $\rightarrow$ rejected by Related work positioning.

Limitations: no real CinC/SHHS numbers in this draft; channel-aware pooling beyond lead zeroing is partial; night `ahi\_bin` uses apnea-epoch-rate cut-points as placeholders; transfer claims bounded to evaluated montages once data exist; strong public baselines (CIMSleepNet, FOCAL ports, full SleepBench) not yet run locally.

Methods (reproducibility)

- Package: ``sleepfm/`` (Python). Configs: ``configs/unify.yaml``, ``configs/unify_temporal.yaml``.
- Figures: ``scripts/plot_unify_figures.py`` with SciencePlots + Times New Roman / SimHei.
- Tests: ``scripts/run_tests.py``, ``scripts/smoke_test.py``.
- Paper suite: ``scripts/run_paper_suite.py --demo`` (synthetic) or ``--data-dir`` after export; paper few-shot defaults to 10 repeats; ``--space-probe`` for shared/private/concat.
- Space probe CLI: ``scripts/eval_space_probe.py``.
- Protocol inventory: ``scripts/protocol_checklist.py``; access steps: ``docs/DATA_ACCESS.md``.
- Compute note (demo): CPU or single GPU; synthetic few-epoch pretrain completes in minutes on a workstation. Full CinC/SHHS wall-clock ****待补充****.

Code / data availability

Public code/docs snapshot:

<https://github.com/Coucou2016/SleepFM-Unify> (``main``; large ``data/`` and ``outputs/`` excluded). PhysioNet/NSRR downloads require user accounts and DUAs. Synthetic and fixture datasets are generated in-repo. License: MIT (see ``LICENSE``); cite via ``CITATION.cff``.

Figures

- ****Fig. 1**** Architecture (missing-PSG stack + shared-private tool heads). Schematic; no clinical metrics.
- ****Fig. 2**** Synthetic Unify pretrain loss curves (demo only).
- ****Fig. 3**** Ablation schematic with chance baseline (demo numbers only; CinC/SHHS 待补充).
- ****Fig. 4**** Shared×private Gram diagnostic (demo forward pass).
- ****Fig. 5**** Modality-dropout robustness schematic (demo; 待补充 on real data).
- ****Fig. 6**** Experiment pipeline from raw PSG to L00 vs Unify evaluation.

References (selected)

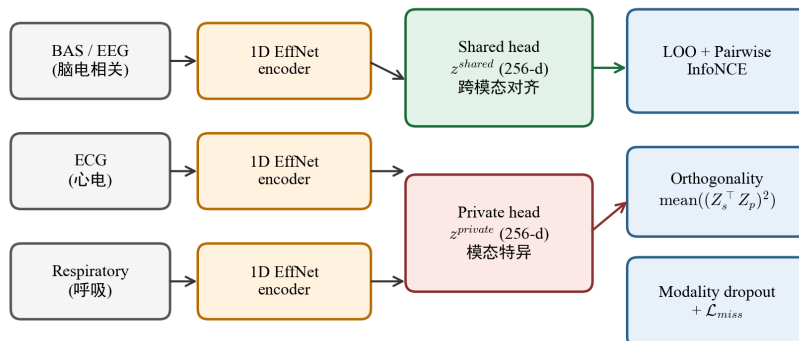
1. Thapa R. et al. SleepFM: Multi-modal Representation Learning for Sleep Across Brain Activity, ECG and Respiratory Signals. *ICML* 2024; PMLR 235:48019–48037; arXiv:2405.17766.
2. A multimodal sleep foundation model for disease prediction. *Nature Medicine* (doi:10.1038/s41591-025-04133-4). (Distinct scale/task from [1]; do not conflate metrics.)
3. CIMSleeNet — Robust Sleep Staging over Incomplete Multimodal Physiological Signals via Contrastive Imagination. *NeurIPS* 2024; <https://github.com/SQAIYY/CIMSleeNet>.
4. Hou et al. Omni-Sleep: A Sleep Foundation Model via Hierarchical Contrastive Learning of CNS – ANS Dynamics. arXiv:2607.07720.
5. FOCAL and shared-private multimodal factorization relatives (shared/private + orthogonality for multimodal time series; see literature notes in ``docs/reports/``).
6. OSF / SleepBench-style sleep foundation benchmarking efforts (heterogeneous evaluation; not replaced here).

Assumptions / missing inputs

- Real CinC/SHHS/MESA metrics: ****待补充**** (no credentials / raw downloads on this host as of 2026-09-14).
- Clinical AHI: not computed; use `apnea\_epoch\_rate` wording.
- Strong external baselines (CIMSsleepNet, FOCAL ports, SleepBench): ****待补充****.

Figures (SciencePlots)

SleepFM-Unify: Shared-Private Factorization (架构示意)
schematic only — no CinC/SHHS metrics



Baseline SleepFM: unify.enabled=false, leave-one-out only. CJK font=SimHei

图1 / Fig. 1. SleepFM-Unify 共享 - 私有架构示意

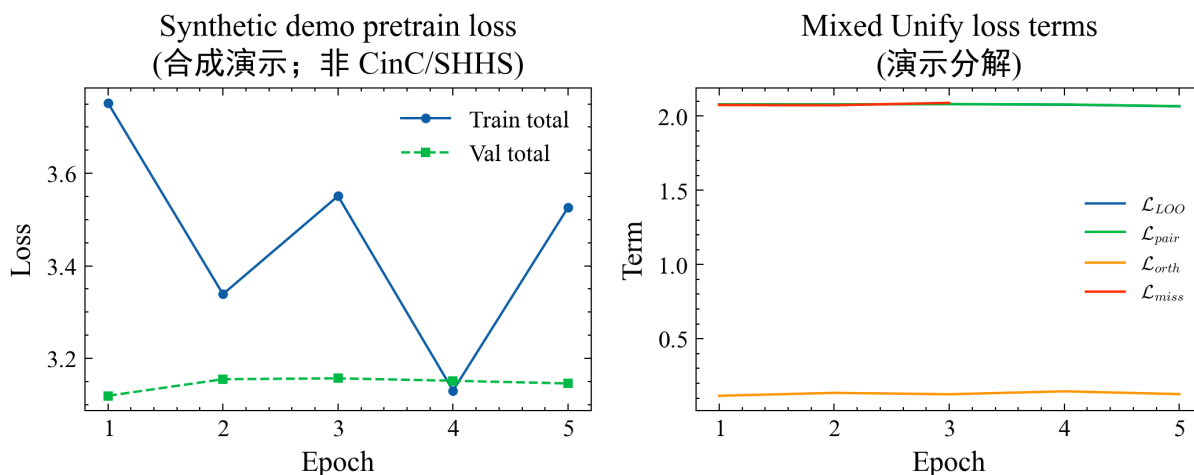


图2 / Fig. 2. 合成演示 Unify 预训练损失曲线

Ablation schematic on synthetic labels
(合成演示≈随机; CinC/SHHS 数值待补充)

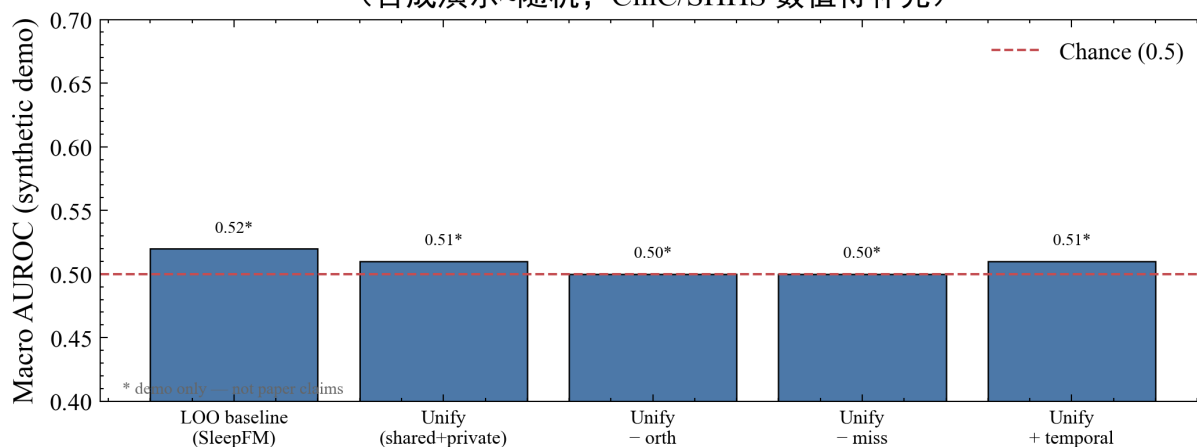


图3 / Fig. 3. 消融示意 (合成≈随机, 非论文主张)

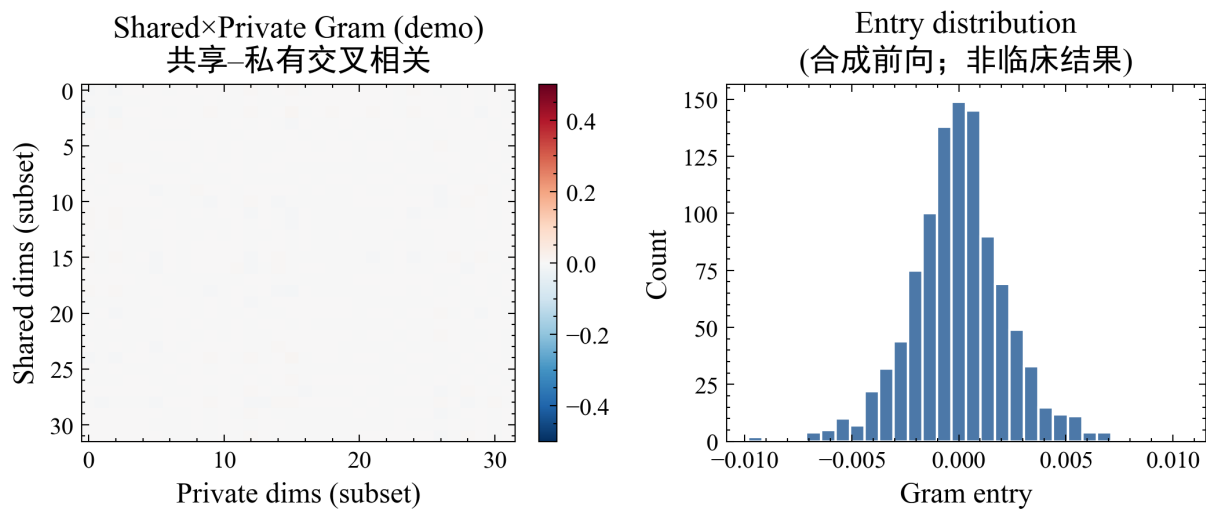


图4 / Fig. 4. 共享×私有 Gram 诊断 (合成前向)

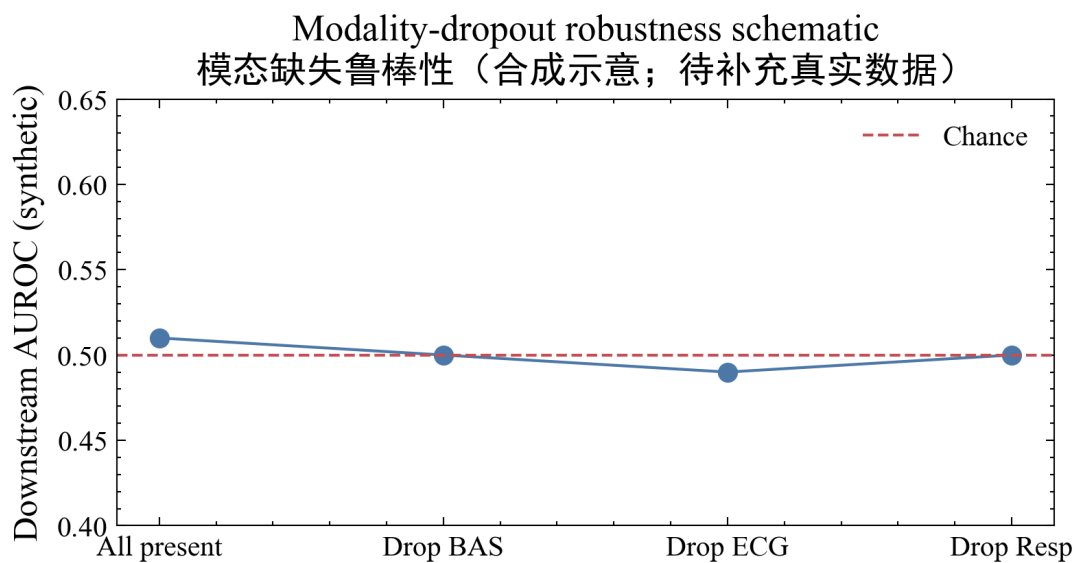


图5 / Fig. 5. 模态缺失鲁棒性示意 (合成)

Experiment pipeline / 实验流水线

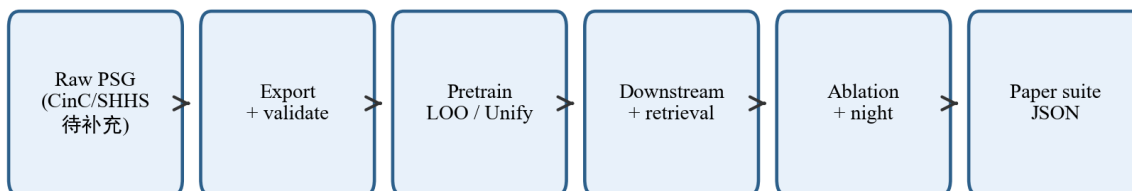


图6 / Fig. 6. 实验流水线